

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461132.772 +/- 0.047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.044
Transit depth (Rp/Rs)^2	2.11 +/- 1.29 [%]
Orbital Inclination (inc)	84.57 +/- 2.55
Airmass coefficient 1 (a1)	1.254 +/- 0.014
Airmass coefficient 2 (a2)	-0.1074 +/- 0.0091
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	None
Optimal Aperture	3.19
Optimal Annulus	8.45
Transit Duration (day)	0.051 +/- 0.043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.044 vs 0.1276 (deviation 0.39σ)
Duration fit vs published	0.051 d vs 0.1179 d (deviation 1.53σ)
Mid-time O–C	45.7 min
Depth SNR	3.2
Residual scatter	1.13 %

Light curve

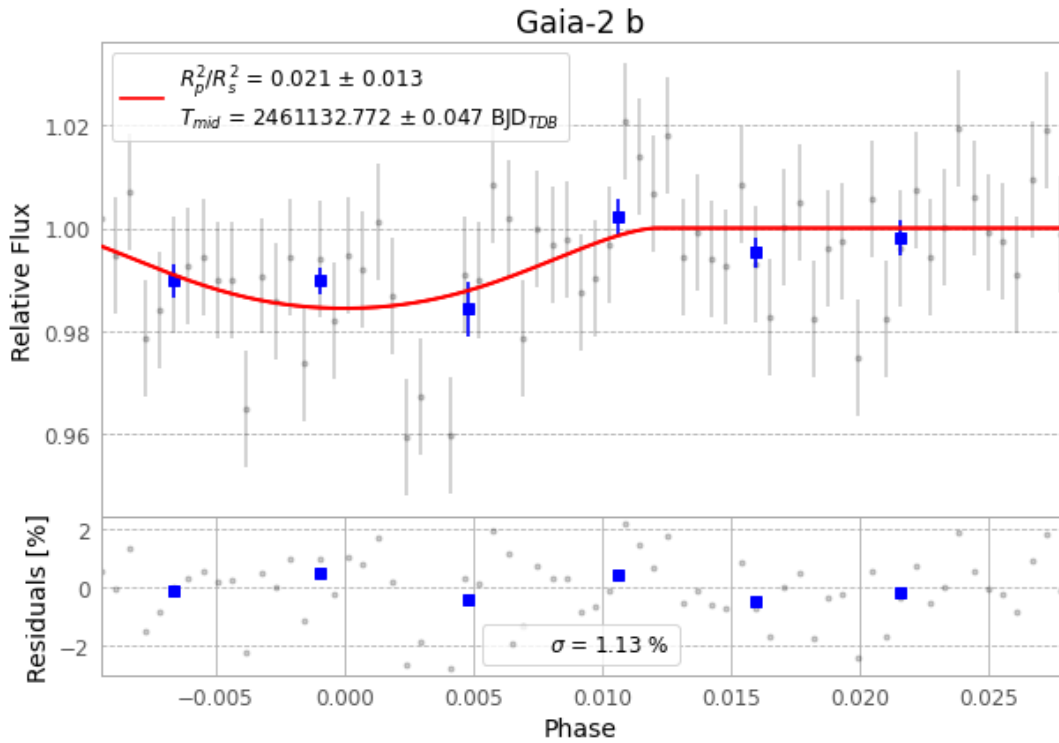


Figure 1 — FinalLightCurve_Gaia-2 b_2026-04-01.png (SHA-256 b0effa22323fa092...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [354, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 921a6249ba3b5220...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

67 FITS frames (44 MB), Gaia-2260402053915.FITS → Gaia-2260402085715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-260402.log (f4fa296819c8...), FinalLightCurve_Gaia-2 b_2026-04-01.png (b0effa22323f...), FinalLightCurve_Gaia-2 b_2026-04-01.csv (72599f92ee04...)

Compute resources

Wall time 136.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 22:49Z.